# A New Technique for the Subdomain Method in Predicting Electromagnetic Performance of Surface-Mounted Permanent Magnet Motors With Shaped Magnets and a Quasi-Regular Polygon Rotor Core

Chuntao Chen, Xinzhen Wu, Xibo Yuan, *Senior Member, IEEE*, and Xiaoqin Zheng, *Member, IEEE*

***Abstract*—Surface-mounted permanent magnet (SPM) motors with shaped magnets and a quasi-regular polygon rotor core (QPRC) are challenging to analyze in terms of accuracy and speed because of the special structure. This article proposes a new technique for the subdomain method which can accurately predict the electromagnetic performance of this kind of motor. The proposed new technique deals with the shaped magnet based on the method of magnet symmetrical segmentation, and it can solve the magnetic field generated by a pair of symmetrical segments together at each time. Especially, for the QPRC, the idea of variable rotor core radius is used in the new technique to analyze different segments. Furthermore, the new technique adopts a periodic boundary condition, which can quickly predict the electromagnetic performance of the motor, such as air-gap flux density, cogging torque, no-load back-EMF, winding inductance, electromagnetic torque and unbalanced magnetic force (UMF). This proposed new technique is applied to a 12-pole/3-phase SPM motor with shaped magnets and a QPRC, and its correctness is verified by finite-element analysis and experiment. The new technique can analyze the motor accurately, and its calculation speed is two orders of magnitude faster than that of the finite-element method.**

***Index Terms*—Magnet symmetrical segmentation, periodic boundary condition, quasi-regular polygon rotor core, shaped magnet, subdomain model, surface-mounted permanent magnet motor.**

## I. Introduction

IN THE past decade, permanent magnet (PM) motors have developed rapidly because of their superior performance, and they are widely used in various applications, such as electric vehicles (EVs), ship propulsion and wind power generation [1], [2], [3]. There are generally two main types of PM motors, i.e., surface-mounted PM (SPM) motors and interior ones that are mainly used in EVs [4]. The SPM motor has the advantages of simple rotor core structures, good sinusoidal characteristics of the air-gap magnetic field and less magnetic leakage, which are very important in applications such as ship electric propulsion [5]. Particularly, the good sinusoidal characteristics of the air-gap magnetic field can reduce the vibration and noise of propulsion motors used in ships. The rotor structure of SPM motors can be divided into the tile type and the arcuate type, as shown in Fig. 1. It can be seen that the rotor of the tile SPM motor is composed of the tile PM and the circular rotor core, while the other consists of the arcuate PM and the quasi-regular polygon rotor core (QPRC). The arcuate PM has a simple structure that has the advantages of convenient processing and installation. In addition, compared with tile SPM motors, arcuate SPM motors can improve the utilization rate of the rotor core, reduce the flux leakage between poles and improve the power density of the motor. Therefore, the arcuate SPM motors are widely used in high-capacity direct-drive PM propulsion systems [6].

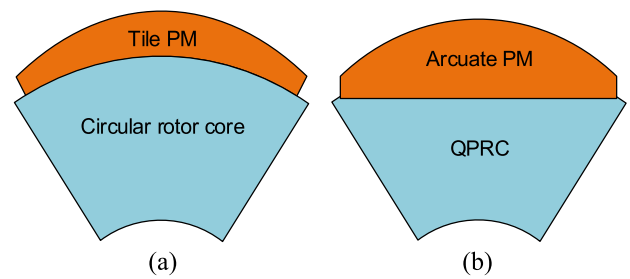


Fig. 1. Rotor structures of the SPM motor. (a) Tile PM and circular rotor core. (b) Arcuate PM and QPRC.

For SPM motors, the magnet-shaping technique effectively improves the sinusoidal characteristics of the air-gap magnetic field and reduces the cogging torque [7]. The common magnet-shaping forms mainly include eccentric shape [8], sinusoidal

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shape [9] and loaf shape [10]. As shown in Fig. 1(a) and (b), both the upper arcs of the PMs use an eccentric shape. Although the magnet-shaping technique has the advantages mentioned above, it causes the PM to be irregular (the PM is of unequal thickness), and the structure of the QPRC also makes the PM irregular. As a result, the difficulty of calculating and analyzing this kind of PMs is more serious compared with regular PMs. The electromagnetic performance of the motors with an irregular PM can be analyzed using finite-element (FE) analysis (FEA). However, its slow calculation speed and large computing resource requirements are unsuitable for it to be used in the initial design and optimization of motors with irregular magnets [11], [12].

The equivalent surface current method and the magnet segmentation method can be used to address the difficulty of calculating the irregular PMs. The equivalent surface current method treats the PM as current sheets [13]. In [14], the PM was converted to current sheets by using the equivalent surface current method, and the actual air-gap magnetic field generated by the current sheets can be obtained by the conformal mapping method combined with the Hague equation [15]. Although the equivalent surface current method can deal with the PM conveniently, it cannot consider the actual permeability of the PM, which is also its shortcoming.

The magnet segmentation method divides the magnet into finite regular segments along the radial direction, and each segment is treated with the same thickness. Then, the air-gap magnetic field generated by the PMs can be calculated using the subdomain method combined with the magnetization vector method [16], [17], [18], [19]. The advantage of the magnet segmentation method is that the actual permeability of the PM can be considered, which can ensure the calculation accuracy of the subdomain model. An analytical model accounting for shaped magnets was developed in [10], where it circumferentially segmented the shaped magnets into finite pieces. The air-gap flux density can be predicted accurately by using the subdomain method. However, the angle between the magnetization direction of the PM and the radial direction must be considered, which is the disadvantage of this analytical model. In [20], a SPM motor with loaf-shape magnets and an octagon rotor was analyzed. The magnet segmentation method combined with the subdomain method was used to calculate the air-gap magnetic density, and the variable radii of the octagon rotor core were considered. However, the variable radii of the rotor core need to be considered separately under the loaded condition, which increases the difficulty of analysis. Furthermore, the analytical models in [10] and [20] can only calculate one segment at a time, which has a slow calculation speed.

In this article, a new technique for the subdomain method is proposed to address the complex rotor structures of SPM motors, including the shaped magnets and the QPRC, as shown in Figs. 1(b) and 2. In the new technique, the $d$-axis of the PM is taken as the centerline, and the magnet is symmetrically split into finite regular arc-shaped segments along the radial direction. Compared with [10] and [20], the proposed new technique can analyze a pair of segments symmetrical to the $d$-axis of the PM in each calculation for faster analysis, which is one of the main contributions of this article. As shown in Fig. 4, because the lower arc radius of each PM segment is different, the outer radius of the rotor core in the subdomain model is changed constantly when calculating different segments, which is a new idea proposed in this article to address the arcuate magnets. Different from [20], the proposed idea of variable rotor core radius can be used not only under the no-load condition but also under the loaded condition. Finally, the total air-gap magnetic density can be obtained by superimposing the air-gap magnetic density generated by all segments. A 12-pole/3-phase eccentric-shaping SPM machine with a QPRC is modeled by using the proposed new technique, and various electromagnetic performances, such as cogging torque, no-load back-EMF, winding inductance, electromagnetic torque and unbalanced magnetic force (UMF), are predicted. Integral and fractional slot motors are established to verify the applicability of the proposed new technique, and its correctness is verified by the FEA and experiments. Furthermore, the new technique adopts the odd or even periodic boundary condition, i.e., the partial motor model is calculated instead of the whole model. Therefore, the calculation speed of the new technique can be significantly improved with high accuracy.

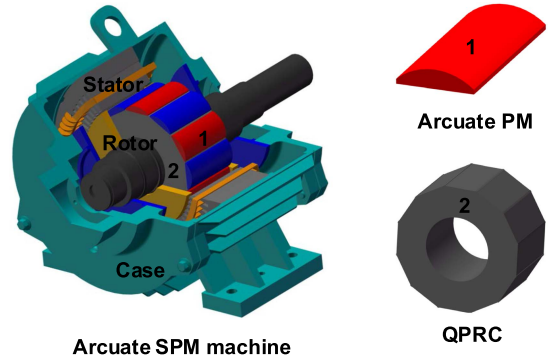


Fig. 2. Structure diagram of the arcuate SPM machine researched in this article.

## II. Motor Topology and Magnet Segmentation

### A. Motor Topology

Fig. 2 shows the structure diagram of the arcuate SPM machine to be analyzed in this article, which mainly includes the case, the stator and the rotor. The rotor of the machine mainly consists of arcuate PMs and the QPRC (quasi-regular dodecagon), and Fig. 1(b) is their cross-sectional view under one pole. Additionally, in this article, the PM is adopted parallel magnetization, and its upper surface is adopted an eccentric shape. The stator is adopted the parallel opening slot.

Table I shows the main parameters of the arcuate SPM motor, and materials of the stator core, the QPRC and arcuate PMs are 35WW300, Q345 and SmCo30, respectively. This motor adopts double-layer short-pitch distributed windings, in which the ratio of the short-pitch $y$ is 5/6, and the number of slots per pole per phase $q$ is 2. The number of series turns per phase $N_1$ is 144. The winding connection type of this motor is the Y connection.

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TABLE I
MAIN PARAMETERS OF THE ARCUATE SPM MOTOR

| Parameter | Value | Parameter | Value |
|---|---|---|---|
| Number of phases | 3 | Stator outer radius | 145 mm |
| Rated power | 10 kW | Stator inner radius | 90 mm |
| Rated speed | 900 r/min | Rotor outer radius | 86.8 mm |
| Number of pole pairs | 6 | Rotor inner radius | 40 mm |
| Number of slots | 72 | Active axial length | 88 mm |
| Rated current (RMS) | 15.9 A | Slot depth | 33 mm |
| PM remanence | 1.065 T | Slot width | 4.2 mm |

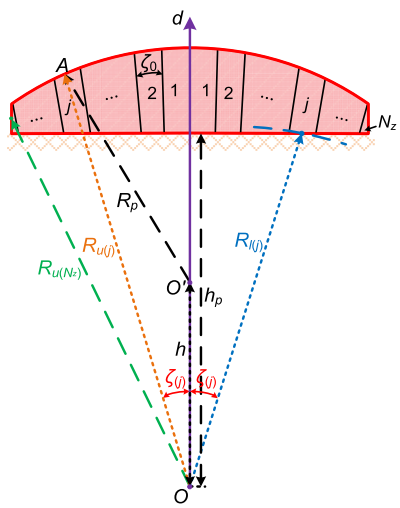


Fig. 3. Diagram of the arcuate magnet segmentation model.

This article takes a high-power propulsion motor used in ships as the research object, and the motor analyzed in this article is a down-scaled prototype of the high-power motor. Hence, the structure of this prototype is basically consistent with the actual high-capacity one.

## B. Magnet Segmentation Method

The magnet segmentation method addresses the motor with the eccentric-shaping arcuate PMs, and Fig. 3 is the arcuate magnet segmentation model.

In Fig. 3, the coordinate origin $O$ is the axis of the motor, and the point $O'$ is the center of the upper arc in the PM. $h$ (57 mm) is the eccentric distance, $R_p$ (29.8 mm) is the eccentric radius, and $h_p$ (77.05 mm) is the distance from the lower surface of the PM to the point $O$. Taking the $d$-axis of the PM as a centerline, the PM is divided into $N_z$ pairs segments along the radial direction (the number of segments under the half pole is $N_z$, numbered 1, 2, …, $j$, …,$N_z$). As can be seen from Fig. 3, the first $N_z$-1 pairs are divided by equal radians, and the radian of each segment $\zeta_0$ can be expressed as

$$\zeta_0 = \frac{\alpha\pi}{2p\,(N_z - 1)} \tag{1}$$

where $\alpha$ (0.922) is the pole-arc to the pole-pitch ratio of the upper arc in the PM, and $p$ is the number of pole pairs.

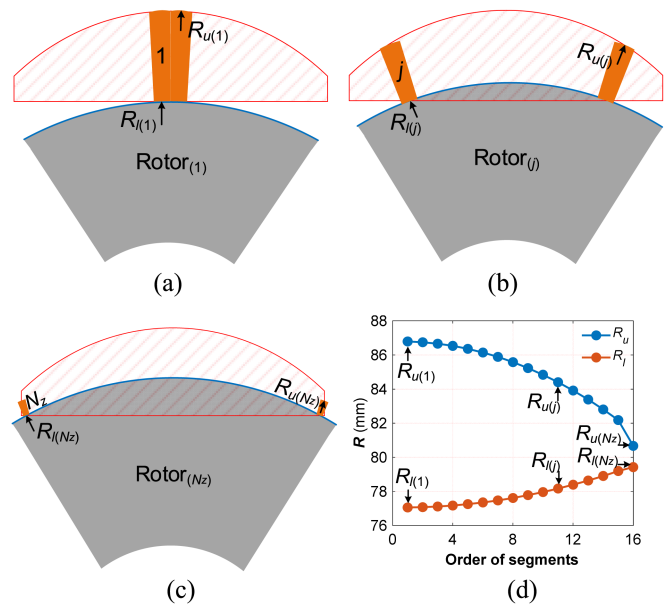


Fig. 4. Variation of $R_u$ and $R_l$ of segments in the PM. (a) First pair of segments. (b) $j$th pair of segments. (c) $N_z$th pair of segments. (d) Curves of $R_u$ and $R_l$.

Then, the angle $\zeta_{(j)}$ between the centerline of the $j$th segment ($j = 1, 2, \ldots, N_z$-1) and the $d$-axis of the PM can be given by

$$\zeta_{(j)} = \frac{(2j-1)\,\zeta_0}{2} \tag{2}$$

If the number of segments is enough, each segment can be regarded as a regular radial arc-shaped magnet. In the triangle $OO'A$, the expression of the upper arc radius $R_{u(j)}$ in the $j$th segment can be obtained according to the Law of Cosines.

$$R_{u(j)}^2 + h^2 - 2hR_{u(j)}\cos(\zeta_{(j)}) = R_p^2 \tag{3}$$

The expression of the lower arc radius $R_{l(j)}$ in the $j$th segment can be written as

$$R_{l(j)} = \frac{h_p}{\cos(\zeta_{(j)})} \tag{4}$$

It can be seen from (3) and (4) that $R_{u(j)}$ and $R_{l(j)}$ will be constantly changed with the change of angle $\zeta_{(j)}$, this is the special issue to be addressed in this article.

In order to ensure the accuracy of the calculation, the $N_z$th pair of segments are also analyzed and equivalent to regular radial arc-shaped segments. The upper arc radius $R_{u(Nz)}$ is the distance from the center of the left (right) side of the PM to the point $O$. The length of the left (right) side in the PM is 2.42 mm, $R_{u(Nz)}$ and $R_{l(Nz)}$ can be obtained easily according to the geometric relationship in Fig. 3.

Fig. 4 shows the variation of the upper arc radius $R_u$ and the lower arc radius $R_l$ of segments. Fig. 4(a), (b) and (c) also show the outer radius of the rotor core corresponding to the first pair of segments, the $j$th pair of segments and the $N_z$th pair of segments, respectively. It can be seen that the outer radius of the rotor core increases in turn. Hence, in the proposed new technique, the outer radius of the rotor core is changed when different segments are analyzed. Fig. 4(d) shows curves of $R_u$

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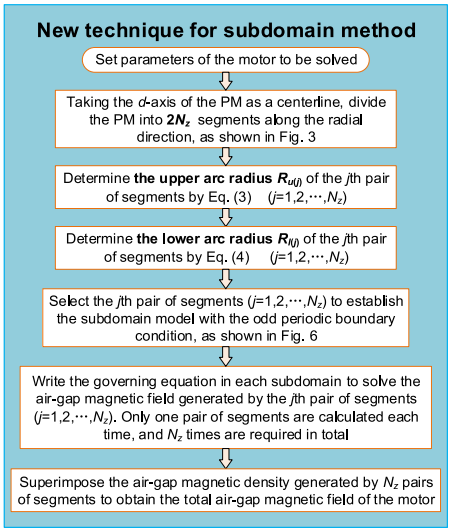


Fig. 5. Flowchart of the new technique for subdomain method.

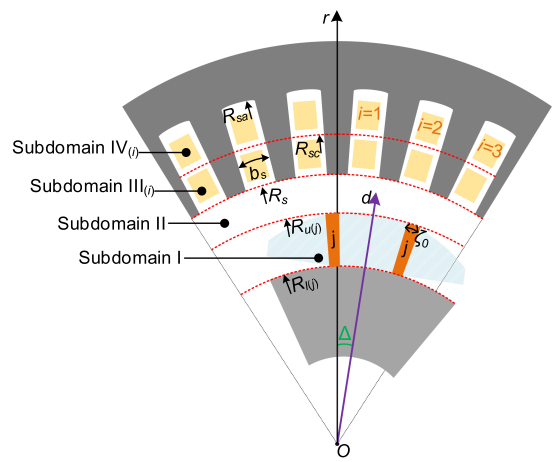


Fig. 6. Subdomain model with the $j$th pair of segments.

and $R_l$ when $N_z = 16$, and the value of $R_l$ increases with the increment of the segment order, but that of $R_u$ is opposite.

Since the proposed new technique can only calculate one pair of segments each time, the more segments in the PM, the more computing time required for the proposed new technique, but the more accurate the calculation results will be.

## III. Solution of Air-Gap Magnetic Field Using New Technique for Subdomain Method

### A. Process of the Proposed New Technique

Fig. 5 shows the flowchart of the proposed new technique. In this new technique, it is a significant step to divide the PM into regular segments, which has been introduced in detail in Section II. The magnetization vector method is used for analyzing the PM when solving the PM subdomain, and a pair of symmetrical segments can be calculated together at each time. Then, the total air-gap magnetic field can be obtained by the superposition of the magnetic field generated by all segments.

In the early stage of the motor design, the proposed new technique is required to calculate a large number of design options. In order to improve the speed and efficiency of the preliminary design optimization, the new technique usually neglects the magnetic saturation effect of the stator (rotor) core, which may reduce the accuracy of the new technique. Then, the FEA considering the magnetic saturation effect is often used afterwards for the verification of the final design scheme.

### B. Calculation of the Air-Gap Magnetic Field

*1) Establishment of the Subdomain Model:* Taking the $j$th pair of segments as an example, a subdomain model is established, as shown in Fig. 6. It can be seen that the motor investigated in this article is divided into four subdomains: the PM subdomain (Subdomain I), the air-gap subdomain (Subdomain II), the upper half slot subdomain (Subdomain $\mathrm{III}_{(i)}$, $i = 1, 2,...,$ $Z/2/p$. $Z$ is the number of slots.) and the lower half slot subdomain (Subdomain $\mathrm{IV}_{(i)}$).

In Fig. 6, the polar coordinate is established, and clockwise is the positive direction. The point $O$ which is the same as the one in Fig. 3 is the coordinate origin, and the centerline of the left tooth of the first slot is the polar axis "$r$". Furthermore, the "$d$" axis is not only the centerline of the PM but also the symmetric axis of the $j$th pair of segments. $R_{l(j)}$ and $R_{u(j)}$ are the same as the ones in Fig. 4(b), $R_s$ is the inner radius of the stator, $R_{sc}$ is the distance from the middle of the slot to the point $O$, $R_{sa}$ is the distance from the slot bottom to the point $O$, and $b_s$ is the slot width. $\Delta$ is the angle between the "$d$" axis and the polar axis "$r$", and it also determines the initial position of the rotor.

In order to simplify the analysis, there are several assumptions for the proposed new technique: (1) The end effect of the motor is ignored; (2) The relative permeability of the stator (rotor) core is infinite; (3) The current density in slots is distributed evenly, and the slot is radial; (4) The PM has linear properties, and the permeability of the air region in Subdomain I is the same as that of the PM.

*2) Solution of Each Subdomain:* Taking the magnetic vector potential $\boldsymbol{A}$ as the solution variable, Laplace or Poisson equation is written in each subdomain, and their solutions can be obtained by using the technique of variable separation. In the two-dimensional field, the magnetic vector potential $\boldsymbol{A}$ has only a $z$-component. The formulas in (5) can simplify expressions of the magnetic vector potential in each subdomain and improve the calculation speed of the proposed new technique [21].

$$P_z(x,y) = \left({}^{x}/_{y}\right)^z + \left({}^{y}/_{x}\right)^z$$
$$E_z(x,y) = \left({}^{x}/_{y}\right)^z - \left({}^{y}/_{x}\right)^z \tag{5}$$

Besides, the periodic boundary condition is also applied to improve the calculation speed. If the greatest common divisor (GCD) of the number of pole pairs $p$ and the number of slots $Z$ equals $c$, the machine rotationally repeats $c$ times of the same structure [22]. If $Z/c$ is even, then the odd periodic boundary condition can be used in the subdomain model, i.e., only $1/2/c$

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partial model is required to analyze. Otherwise, the even periodic boundary condition is used (1/$c$ partial model is analyzed).

For the motor analyzed in this article, $Z/c$ is even ($c$ equals the number of pole pairs $p$ in this article), so the odd periodic boundary condition can be used, and the number of slots in the subdomain model is reduced from $Z$ to $Z/2/c$. The magnetic vector potential in Subdomain I and Subdomain II can satisfy the following odd periodic boundary condition.

$$A_z(r,\theta) = -A_z\left(r, \theta + \frac{\pi}{c}\right) \tag{6}$$

where $r$ is the radius variable of $A_z$, and $\theta$ is the angle variable (mechanical angle).

*a) Solution in Subdomain I:* The magnetization vector method is used to analyze the PM. Since the PM researched in this article is adopted parallel magnetization, the radial component $M_{r(j)}$ and the tangential component $M_{\theta(j)}$ of the magnetization vector of the $j$th segment in the range of $0\sim\pi/2p$ (the half pole) can be expressed as

$$M_{r(j)}(\theta) = \begin{cases} 0 & 0 \le \theta - \Delta \le (j-1)\zeta_0 \\ \frac{B_r\cos(\theta-\Delta)}{\mu_0} & (j-1)\zeta_0 \le \theta - \Delta \le j\zeta_0 \\ 0 & j\zeta_0 \le \theta - \Delta \le \pi/2p \end{cases}$$

$$j = 1, 2, 3\ldots \tag{7}$$

$$M_{\theta(j)}(\theta) = \begin{cases} 0 & 0 \le \theta - \Delta \le (j-1)\zeta_0 \\ -\frac{B_r\sin(\theta-\Delta)}{\mu_0} & (j-1)\zeta_0 \le \theta - \Delta \le j\zeta_0 \\ 0 & j\zeta_0 \le \theta - \Delta \le \pi/2p \end{cases}$$

$$j = 1, 2, 3\ldots \tag{8}$$

where $B_r$ is the remanence density of the PM, and $\mu_0$ is the vacuum permeability.

Since $M_{r(j)}(\theta)$ and $M_{\theta(j)}(\theta)$ are even and odd symmetric periodic functions respectively, Fourier decomposition of $M_{r(j)}(\theta)$ and $M_{\theta(j)}(\theta)$ can be carried out in the range of one pole ($0\sim\pi/p$), and the Fourier expansion of $M_{r(j)}$ and $M_{\theta(j)}$ can be obtained as following.

$$M_{r(j)}(\theta) = \sum_{mc/p=1,3,5,\ldots}^{\infty} M_{r(j)m}\cos[mc(\theta-\Delta)]$$

$$M_{\theta(j)}(\theta) = \sum_{mc/p=1,3,5,\ldots}^{\infty} M_{\theta(j)m}\sin[mc(\theta-\Delta)] \tag{9}$$

where $m$ is the order of harmonics in Subdomain I.

When $mc\neq 1$

$$M_{r(j)m} = \frac{2pB_r}{\pi\mu_0}(K_a + K_b)$$

$$M_{\theta(j)m} = \frac{2pB_r}{\pi\mu_0}(K_a - K_b) \tag{10}$$

where

$$K_a = \frac{\sin[(mc+1)j\zeta_0] - \sin[(mc+1)(j-1)\zeta_0]}{mc+1}$$

$$K_b = \frac{\sin[(mc-1)j\zeta_0] - \sin[(mc-1)(j-1)\zeta_0]}{mc-1} \tag{11}$$

When $mc = 1$

$$M_{r(j)1} = M_{\theta(j)1} = \frac{pB_r}{\pi\mu_0}\{\sin(2j\zeta_0) - \sin[2(j-1)\zeta_0]\} \tag{12}$$

In (9)–(12), the magnetic field generated by the $j$th pair of segments under all poles can be calculated at once. Therefore, the number of poles does not affect the number of subdomains in the model.

What's more, when the PMs adopt radial magnetization, the radial component $M_{r(j)}$ and the tangential component $M_{\theta(j)}$ of the magnetization vector of the $j$th segment in the range of $0\sim\pi/2p$ (the half pole) can be expressed as

$$M_{r(j)}(\theta) = \begin{cases} 0 & 0 \le \theta - \Delta \le (j-1)\zeta_0 \\ \frac{B_r}{\mu_0} & (j-1)\zeta_0 \le \theta - \Delta \le j\zeta_0 \\ 0 & j\zeta_0 \le \theta - \Delta \le \pi/2p \end{cases}$$

$$j = 1, 2, 3\ldots \tag{13}$$

$$M_{\theta(j)}(\theta) = 0 \qquad 0 \le \theta - \Delta \le \pi/2p$$

$$j = 1, 2, 3\ldots \tag{14}$$

According to (9), (13) and (14) can also be Fourier decomposed in the range of $0 \sim \pi/p$, and the expressions of $M_{r(j)m}$ and $M_{\theta(j)m}$ can be calculated as

$$M_{r(j)m} = \frac{4B_r}{m\pi\mu_0}\{\sin[mc(j\zeta_0)] - \sin[mc(j-1)\zeta_0]\}$$

$$M_{\theta(j)m} = 0 \tag{15}$$

Next, the magnetic vector potential $A^j{}_{z1}$ in Subdomain I can satisfy the Poisson equation, which is shown in (16).

$$\frac{\partial^2 A^j_{z1}}{\partial r^2} + \frac{1}{r}\frac{\partial A^j_{z1}}{\partial r} + \frac{1}{r^2}\frac{\partial^2 A^j_{z1}}{\partial\theta^2} = -\frac{\mu_0}{r}\left(M_{\theta(j)} - \frac{\partial M_{r(j)}}{\partial\theta}\right)$$

$$R_{l(j)} \le r \le R_{u(j)} \tag{16}$$

According to the method of variable separation, the magnetic vector potential $A^j{}_{z1}$ can be given by

$$\begin{aligned} A^j_{z1} = &\sum_{m=1}^{\infty}\left[A_{m1}\frac{P_{mc}(r, R_{l(j)})}{P_{mc}(R_{u(j)}, R_{l(j)})}\right. \\ &\left. - X_{mj}(r)\sin(mc\Delta)\right]\cos(mc\theta) \\ &+ \sum_{m=1}^{\infty}\left[C_{m1}\frac{P_{mc}(r, R_{l(j)})}{P_{mc}(R_{u(j)}, R_{l(j)})}\right. \\ &\left. + X_{mj}(r)\cos(mc\Delta)\right]\sin(mc\theta) \end{aligned}$$

$$R_{l(j)} \le r \le R_{u(j)}, \quad 0 \le \theta \le \pi/c$$

$$m = 1, 3, 5\ldots \tag{17}$$

It is worth noting that the odd periodic boundary condition is mainly used in this article, the order of harmonics $m$ is equal to

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1,3,5 …, if *Z*/*c* is odd, i.e., the even periodic boundary condition should be used in the subdomain model, the order of harmonics *m* is equal to 1,2,3 …, and (17) is rewritten as

$$\begin{aligned} A_{z1}^{j} = & \sum_{m=1}^{\infty}\left[A_{m1}\frac{P_{mc}(r,R_{l(j)})}{P_{mc}(R_{u(j)},R_{l(j)})}\right. \\ & \left. - X_{mj}(r)\sin(mc\Delta)\right]\cos(mc\theta) \\ & + \sum_{m=1}^{\infty}\left[C_{m1}\frac{P_{mc}(r,R_{l(j)})}{P_{mc}(R_{u(j)},R_{l(j)})}\right. \\ & \left. + X_{mj}(r)\cos(mc\Delta)\right]\sin(mc\theta) \\ & R_{l(j)} \le r \le R_{u(j)}, \quad 0 \le \theta \le 2\pi/c \\ & m = 1,2,3\ldots \end{aligned} \tag{18}$$

$$\begin{aligned} X_{mj}(r) = & \frac{R_{l(j)}}{mc}\left(\frac{R_{l(j)}}{r}\right)^{mc}\left[k'_{mj}(R_{l(j)})\right. \\ & \left. + \mu_0 M_{\theta(j)m}\right] + k_{mj}(r) \\ & - \frac{P_{mc}\left(r,R_{l(j)}\right)}{P_{mc}\left(R_{u(j)},R_{l(j)}\right)}\left\{\frac{R_{l(j)}}{mc}\left(\frac{R_{l(j)}}{R_{u(j)}}\right)^{mc}\right. \\ & \left. \times\left[k'_{mj}(R_{l(j)}) + \mu_0 M_{\theta(j)m}\right] + k_{mj}(R_{u(j)})\right\} \end{aligned} \tag{19}$$

$$k_{mj}(r) = \begin{cases} \frac{\mu_0(mcM_{r(j)m}+M_{\theta(j)m})r}{(mc)^2-1} & mc \ne 1 \& m = 1,3,5,\ldots \\ -\frac{\mu_0(M_{r(j)1}+M_{\theta(j)1})r\ln r}{2} & mc = 1 \end{cases} \tag{20}$$

where $k'_{mj}(r)$ is the derivative of $k_{mj}(r)$ with respect to $r$, $A_{m1}$ and $C_{m1}$ are unknown coefficients in Subdomain I.

The equation deformation is used in (17), (18) and (19), and it can simplify expressions of the magnetic vector potential at the boundary of each subdomain. For example, in (19), when $r = R_{u(j)}$, $X_{mj}(R_{u(j)}) = 0$.

*b) Solution in Subdomain II:* The Laplace equation is written in Subdomain II, and its solution $A^{j}{}_{z2}$ is shown in (21).

$$\begin{aligned} A_{z2}^{j} = & \sum_{m=1}^{\infty}\left[A_{m2}\frac{R_{u(j)}}{mc}\frac{P_{mc}(r,R_s)}{E_{mc}(R_{u(j)},R_s)}\right. \\ & \left. + B_{m2}\frac{R_s}{mc}\frac{P_{mc}(r,R_{u(j)})}{E_{mc}(R_s,R_{u(j)})}\right]\cos(mc\theta) \\ & + \sum_{m=1}^{\infty}\left[C_{m2}\frac{R_{u(j)}}{mc}\frac{P_{mc}(r,R_s)}{E_{mc}(R_{u(j)},R_s)}\right. \\ & \left. + D_{m2}\frac{R_s}{mc}\frac{P_{mc}(r,R_{u(j)})}{E_{mc}(R_s,R_{u(j)})}\right]\sin(mc\theta) \\ & R_{u(j)} \le r \le R_s, \quad 0 \le \theta \le \pi/c \\ & m = 1,3,5\ldots \end{aligned} \tag{21}$$

Similarly, if *Z*/*c* is odd, (21) is rewritten as

$$\begin{aligned} A_{z2}^{j} = & \sum_{m=1}^{\infty}\left[A_{m2}\frac{R_{u(j)}}{mc}\frac{P_{mc}(r,R_s)}{E_{mc}(R_{u(j)},R_s)}\right. \\ & \left. + B_{m2}\frac{R_s}{mc}\frac{P_{mc}(r,R_{u(j)})}{E_{mc}(R_s,R_{u(j)})}\right]\cos(mc\theta) \\ & + \sum_{m=1}^{\infty}\left[C_{m2}\frac{R_{u(j)}}{mc}\frac{P_{mc}(r,R_s)}{E_{mc}(R_{u(j)},R_s)}\right. \\ & \left. + D_{m2}\frac{R_s}{mc}\frac{P_{mc}(r,R_{u(j)})}{E_{mc}(R_s,R_{u(j)})}\right]\sin(mc\theta) \\ & R_{u(j)} \le r \le R_s, \quad 0 \le \theta \le 2\pi/c \\ & m = 1,2,3\ldots \end{aligned} \tag{22}$$

where $A_{m2}$, $B_{m2}$, $C_{m2}$ and $D_{m2}$ are unknown coefficients in Subdomain II.

*c) Solutions in Subdomain III$_{(i)}$ and Subdomain IV$_{(i)}$:* The number of slots in Subdomain III$_{(i)}$ and Subdomain IV$_{(i)}$ is reduced from Z to Z/2/c by using the odd periodic boundary condition, which can greatly reduce the dimension of the following matrix and improve the calculation speed of the proposed new technique. Poisson equations are written in Subdomain III$_{(i)}$ and Subdomain IV$_{(i)}$, and their solutions $A^{j}{}_{z3i}$ and $A^{j}{}_{z4i}$ are shown in (23) and (24).

$$\begin{aligned} A_{z3i}^{j} = & B_{03i} + \left[\frac{\mu_0 R_{sc}^2(J_{ui}-J_{di})}{2} + \frac{\mu_0 R_{sa}^2}{2}J_{di}\right]\ln\frac{r}{R_s} \\ & + \frac{\mu_0 J_{ui}(R_s^2-r^2)}{4} + \sum_{n=1}^{\infty}A_{n3i}\frac{P_{G_n}(r,R_{sa})}{P_{G_n}(R_s,R_{sa})} \\ & \times\cos\left[G_n\left(\theta+\frac{b_s}{2}-\theta_i\right)\right] \\ & R_s \le r \le R_{sc}, \quad -\frac{b_s}{2}+\theta_i \le \theta \le \frac{b_s}{2}+\theta_i \end{aligned} \tag{23}$$

$$\begin{aligned} A_{z4i}^{j} = & B_{04i} + \frac{\mu_0 J_{di}R_{sa}^2}{2}\ln\frac{r}{R_{sc}} + \frac{\mu_0 J_{di}(R_{sc}^2-r^2)}{4} \\ & + \sum_{n=1}^{\infty}A_{n4i}\frac{P_{G_n}(r,R_{sa})}{P_{G_n}(R_s,R_{sa})}\cos\left[G_n\left(\theta+\frac{b_s}{2}-\theta_i\right)\right] \\ & R_{sc} \le r \le R_{sa}, \quad -\frac{b_s}{2}+\theta_i \le \theta \le \frac{b_s}{2}+\theta_i \end{aligned} \tag{24}$$

Then,

$$G_n = {}^{n\pi}/_{b_s}, \qquad \theta_i = \frac{\pi}{Z}(2i-1),\ i = 1,2,\ldots,{}^{Z}/_{2c} \tag{25}$$

where *n* is the order of harmonics in Subdomain III$_{(i)}$ and Subdomain IV$_{(i)}$. $B_{03i}$, $A_{n3i}$, $B_{04i}$ and $A_{n4i}$ are unknown coefficients to be determined in the above two subdomains. $J_{ui}$ and $J_{di}$ are the current densities in Subdomain III$_{(i)}$ and Subdomain IV$_{(i)}$, respectively. Moreover, $b_s$ is the slot width expressed in radians, and $\theta_i$ is the angle between the centerline of the *i*th slot and the polar axis "*r*".

It is worth mentioning that the motor investigated in this article adopts opening slots, therefore, the effect of the tooth-shoe does not need to be considered.

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*3) Solution of Unknown Coefficients:* There are ten unknown coefficients ($A_{m1}$, $C_{m1}$, $A_{m2}$, $B_{m2}$, $C_{m2}$, $D_{m2}$, $A_{n3i}$, $B_{03i}$, $A_{n4i}$ and $B_{04i}$) in the above four subdomains. Next, the boundary condition at the interface between Subdomain I and Subdomain II is introduced as an example to determine expressions of unknown coefficients.

When $r = R_{u(j)}$,

$$\begin{cases} A_{z1}^{j}\Big|_{r=R_{u(j)}} = A_{z2}^{j}\Big|_{r=R_{u(j)}} & j = 1, 2, 3 \ldots \\ \frac{1}{\mu_r}\frac{\partial A_{z1}^{j}}{\partial r}\Big|_{r=R_{u(j)}} = \frac{\partial A_{z2}^{j}}{\partial r}\Big|_{r=R_{u(j)}} & j = 1, 2, 3 \ldots \end{cases} \tag{26}$$

where $\mu_r$ is the relative permeability of the PM.

It can be seen from (26) that the actual relative permeability $\mu_r$ of the PM can be considered in this article. With the help of Fourier expansion formulas, the equations of the above unknown coefficients can be obtained, which will be shown in the Appendix.

In this article, the equations of unknown coefficients are rewritten into the matrix form [13], and the maximum values of the harmonic order $m$ and $n$ are set to 200 and 10, respectively. Then, the numerical solution of those coefficients can be calculated by using the following formula.

$$\mathbf{X} = \mathbf{K}^{-1}\mathbf{Y} \tag{27}$$

where $\mathbf{X}$ is the column vector of unknown coefficients, such as $A_{m1}$, $C_{m1}$ and so on. $\mathbf{K}$ and $\mathbf{Y}$ are the coefficient matrix and the column vector obtained from the equations of unknown coefficients in the Appendix, respectively.

It is worth noting that the number of slots in the motor will affect the dimension of the coefficient matrix $\mathbf{K}$. The more slots in the motor, the larger the dimension of the matrix $\mathbf{K}$, which will reduce the calculation speed of the proposed new technique. According to (27), the numerical solution of the magnetic vector potential $A^j{}_z\,(r, \theta)$ in each subdomain can be obtained.

*4) Superposition Principle:* According to the solution of the magnetic vector potential in Subdomain II, the air-gap flux density generated by the $j$th pair of segments can be obtained according to following formulas.

$$B_{r(j)}(r,\theta) = \frac{1}{r}\frac{\partial A_{z2}^{j}(r,\theta)}{\partial \theta}, \quad B_{\theta(j)}(r,\theta) = -\frac{\partial A_{z2}^{j}(r,\theta)}{\partial r} \tag{28}$$

where $B_{r(j)}$ and $B_{\theta(j)}$ are radial the tangential components of the air-gap flux density, respectively.

Finally, the total air-gap flux density can be obtained by using the superposition technique, as shown in (29).

$$B_{\mathrm{r}}(r,\theta) = \sum_{j=1}^{N_z} B_{r(j)}(r,\theta), \quad B_{\theta}(r,\theta) = \sum_{j=1}^{N_z} B_{\theta(j)}(r,\theta) \tag{29}$$

where $B_r$ and $B_\theta$ are the radial and tangential components of the total air-gap flux density, respectively. It is worth mentioning that (29) can calculate the no-load or load magnetic field of the motor, which depends on whether there are current densities in stator windings.

## C. Calculation of Electromagnetic Performance

There are various electromagnetic performances of the motor being solved, such as cogging torque, electromagnetic torque, no-load back-EMF, winding inductance and UMF.

*1) Cogging Torque and Electromagnetic Torque:* Using the Maxwell tensor method, the cogging or electromagnetic torque can be obtained, as shown in (30).

$$T = \frac{L_{ef}R_g^2}{\mu_0}\int_0^{2\pi} B_r(R_g,\theta)\cdot B_\theta(R_g,\theta)\mathrm{d}\theta \tag{30}$$

where $L_{ef}$ is the active axial length of the motor, and $R_g$ is the radius in the middle of the air gap, where $R_g = 88.4$ mm in this article.

Under the no-load condition, (30) can calculate the cogging torque of the motor; under the loaded condition, the electromagnetic torque can be calculated.

*2) No-Load Back-EMF and Winding Inductance:* Under the no-load condition, the no-load back-EMF can be obtained by calculating the flux linkage with one-phase winding. The flux linkage with a coil can be calculated by the following formula.

$$\Phi = \sum_{j=1}^{N_z}\frac{L_{ef}N_1}{S}\left[\int_{\theta_i-b_s/2}^{\theta_i+b_s/2}\int_{R_s}^{R_{sc}} A_{z3i}^{j}(r,\theta)r\mathrm{d}r\mathrm{d}\theta - \int_{\theta_{(i+y)}-b_s/2}^{\theta_{(i+y)}+b_s/2}\int_{R_{sc}}^{R_{sa}} A_{z4(i+y)}^{j}(r,\theta)r\mathrm{d}r\mathrm{d}\theta\right] \tag{31}$$

where $N_1$ is the number of turns per coil, $S$ is the area of the half slot, and $y$ is the coil pitch (expressed in the number of slots).

The flux linkage of each phase can be obtained by summing flux linkages associated with all coils under one phase. Then, the no-load back-EMF can be calculated by Faraday's Law [23].

The phase inductance can also be obtained according to the flux linkage method. Taking phase $A$ as an example, when the PM has no remanence density ($B_r = 0$), and DC $I_a$ is applied to phase $A$, the self and mutual inductances of phase $A$ are shown in (32).

$$L_{AA} = \frac{\Phi_A}{I_a}, M_{AB} = \frac{\Phi_B}{I_a}, M_{AC} = \frac{\Phi_C}{I_a} \tag{32}$$

where $L_{AA}$ is the self-inductance of phase $A$, $M_{AB}$ and $M_{AC}$ are mutual-inductances between phase $A$ and $B$, between phase $A$ and $C$, respectively. $\Phi_A$, $\Phi_B$ and $\Phi_C$ are flux linkages of phases $A$, $B$ and $C$, respectively.

*3) Magnetic Local Traction and UMF:* The UMF is caused by an asymmetric magnetic field distribution in the air gap [24]. Firstly, according to the Maxwell tensor method and the loaded air-gap flux density, the loaded magnetic local traction in $x$ and $y$ directions in the middle of the air gap can be calculated according to the following formulas [25].

$$f_x(\theta) = \frac{1}{2\mu_0}\left[B_r^2(R_g,\theta) - B_\theta^2(R_g,\theta)\right]\cos\theta - \frac{1}{\mu_0}B_r(R_g,\theta)\cdot B_\theta(R_g,\theta)\sin\theta \tag{33}$$

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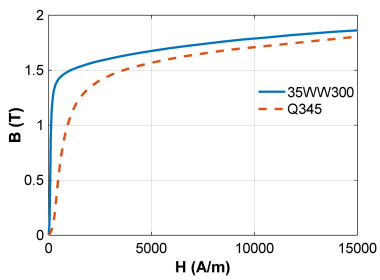


Fig. 7. *BH* curves of 35WW300 (the material of the stator core) and Q345 (the material of the rotor core).

$$f_y(\theta) = \frac{1}{2\mu_0}\left[B_r^2(R_g,\theta) - B_\theta^2(R_g,\theta)\right]\sin\theta + \frac{1}{\mu_0}B_r(R_g,\theta)\cdot B_\theta(R_g,\theta)\cos\theta \tag{34}$$

Then, the components $F_x$ and $F_y$ of the UMF can be expressed as

$$F_x = L_{ef}R_g\int_0^{2\pi} f_x(\theta)\,\mathrm{d}\theta \tag{35}$$

$$F_y = L_{ef}R_g\int_0^{2\pi} f_y(\theta)\,\mathrm{d}\theta \tag{36}$$

The UMF can be obtained according to following formula.

$$F = \sqrt{F_x^2 + F_y^2} \tag{37}$$

## IV. Verification of Accuracy and Rapidity for Proposed New Technique

In this section, the accuracy of the proposed new technique is verified by FEA and experiments. Its applicability is further analyzed by establishing integral and fractional slot motors. What's more, its rapidity can be compared by the subdomain model without the periodic boundary condition and the FEA. In the following calculation, the number of segment pairs $N_z$ is taken as 16.

### A. FE Verification

According to the parameters in Table I, a 1/12 FE model of the arcuate SPM motor investigated in this article is established in a FE software. In the transient field of the FE software, the simulation time is one electric period, i.e., $1/f_N$ s, where $f_N$ is the rated frequency of the motor. The simulation time step is $1/f_N/480$ s. In order to calculate the actual performance of the motor, the material properties and the motor structure in the FEA are consistent with the real motor, and the *BH* curves of stator and rotor core materials are shown in Fig. 7.

Fig. 8 shows the distribution of mesh and rated loaded flux density. It can be seen from Fig. 8(a) that the mesh quality in the FE model is excellent, especially in the air gap. There are 10788 meshes in this FE model, which can ensure the accuracy of FEA. In Fig. 8(b), the rotor rotates at an angle so that the *d*-axis of PM coincides with the axis of phase *A*. Under the loaded condition,

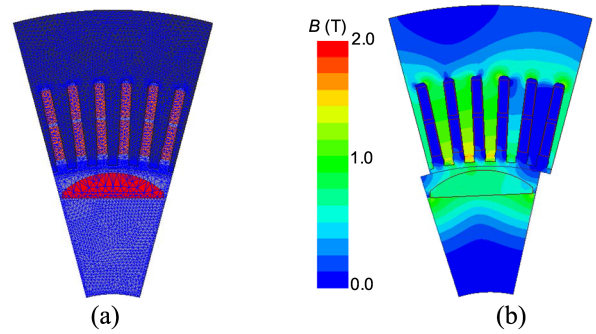


Fig. 8. Field diagram. (a) Mesh. (b) Rated loaded flux density.

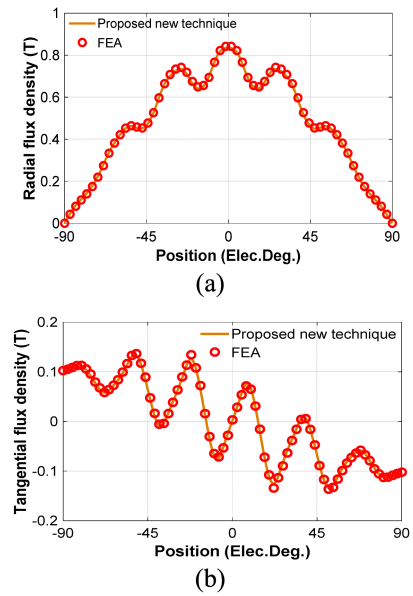


Fig. 9. No-load flux densities under one pole. (a) Radial component. (b) Tangential component.

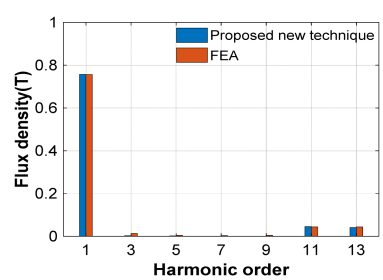


Fig. 10. Harmonic analysis of the no-load radial flux density.

the top of the tooth near the *d*-axis of PM is slightly saturated, but the flux density in the air gap is less than 1T.

*1) No-Load Air-Gap Flux Density:* The no-load radial and tangential flux densities at the middle of the air gap are calculated by using the proposed new technique and the FEA, as can be seen in Fig. 9. Then, the harmonic analysis of the radial component is shown in Fig. 10.

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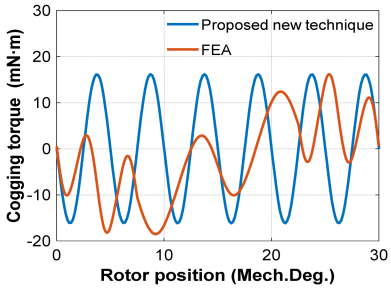


Fig. 11. Cogging torque waveforms predicted by the proposed new technique and the FEA.

TABLE II
COMPARISON OF AMPLITUDES OF THE COGGING TORQUE

| | Proposed new technique | FEA | Error |
|---|---|---|---|
| Amplitude | 16.14 mN·m | 18.51 mN·m | 12.80% |

In Fig. 9, no-load radial and tangential flux densities calculated by the proposed new technique are consistent with the FEA. Furthermore, it can be calculated from Fig. 10 that the difference of fundamental values of the no-load radial flux density between the two methods is only 0.08% when $N_z = 16$. Then, when $N_z$ is taken as 4, 8, 12 and 20, the above differences are 1.24%, 0.63%, 0.24% and 0.07%, respectively. It shows that with the increase of $N_z$, the calculation accuracy of the proposed new technique is gradually improved, but the calculation time is also increasing. Thus, after carefully considering the calculation accuracy and speed, $N_z$ is selected as 16. In addition, the 3rd, 5th and 7th harmonics of the radial component are very small because the PM is adopted the eccentric-shaping technique. The 11th and 13th harmonics are large because of the opening slot investigated in this article.

*2) Cogging Torque:* According to the no-load air-gap flux density, the cogging torque can be obtained using (30). Using the new technique and the FEA, Fig. 11 shows the cogging torque waveforms of the rotor rotating from 0° to 30°, and amplitudes of the cogging torque solved by these two methods are shown in Table II.

According to (38), when the rotor of the proposed motor rotates within one tooth pitch (the mechanical angle is 5°), the cycle number $N_{cog}$ of cogging torque is 1. Then, Fig. 11 shows the cogging torque waveform with six cycles. Because of the effect of the QPRC, the cogging torque waveform is distorted in the FEA, while the result of the proposed new technique is a regular waveform. It can be seen that compared with the FEA which can accurately calculate the waveform of the cogging torque, the new technique has a deviation for calculating the cogging torque because the rotor core corresponding to each pair of segments in the PM is circular rather than a quasi-regular polygon.

$$N_{cog} = \frac{2p}{GCD\,(Z, 2p)} \tag{38}$$

where $GCD(Z, 2p)$ is the maximum common divisor of the number of slots $Z$ and the number of poles $2p$.

TABLE III
COMPARISON OF PHASE $A$ INDUCTANCES

| Inductance | Proposed new technique | FEA | Error |
|---|---|---|---|
| $L_{AA}$ | 6.12 mH | 6.49 mH | 5.70% |
| $M_{AB}$ | -1.58 mH | -1.57 mH | 0.64% |
| $M_{AC}$ | -1.58 mH | -1.57 mH | 0.64% |

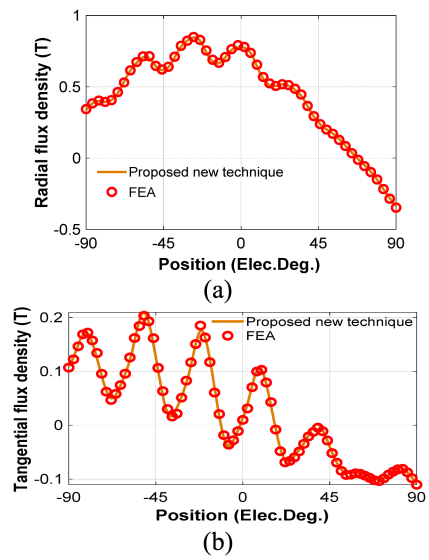


Fig. 12. Loaded flux densities under one pole. (a) Radial component. (b) Tangential component.

It can be seen from Table II that the error of the cogging torque amplitudes calculated by the above two methods is 12.80%, which is a large difference. The eccentric-shaping technique is used for the PM, so the cogging torque of the motor is small (less than 20 mN·m). The smaller cogging torque, the more difficult it is to calculate accurately, but this new technique can estimate the cogging torque in the motor design.

*3) Winding Inductance:* Using (31) and (32), inductances of phase $A$, including self-inductance $L_{AA}$, mutual inductances $M_{AB}$ and $M_{AC}$, are calculated by the proposed new technique, and their results are compared with the FEA, as shown in Table III. It is worth noting that the results of the FEA are the average values.

In Table III, except that the error of $L_{AA}$ is slightly larger, the results of the proposed new technique are basically consistent with the FEA, which shows that the accuracy of this new technique in calculating the winding inductance can be acceptable. In addition, the inductances of phase $B$ and phase $C$ can be calculated in the same way.

*4) Loaded Air-Gap Flux Density:* Under the rated loaded condition, the $I_d = 0$ control mode is used in this article. Fig. 12 shows the rated loaded air-gap flux densities obtained by the proposed new technique and the FEA, and Fig. 13 shows the harmonic analysis of the loaded radial air-gap flux density.

It can be seen from Fig. 12 that the loaded air-gap flux densities obtained by the above two methods are in good agreement with each other. Furthermore, in Fig. 13, the difference of

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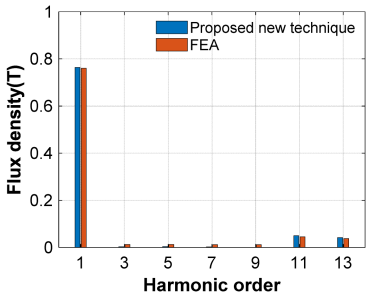


Fig. 13. Harmonic analysis of the loaded radial flux density.

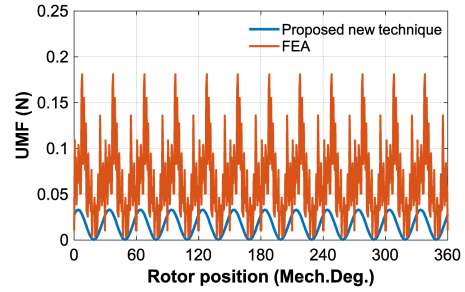


Fig. 15. Comparison of the UMFs.

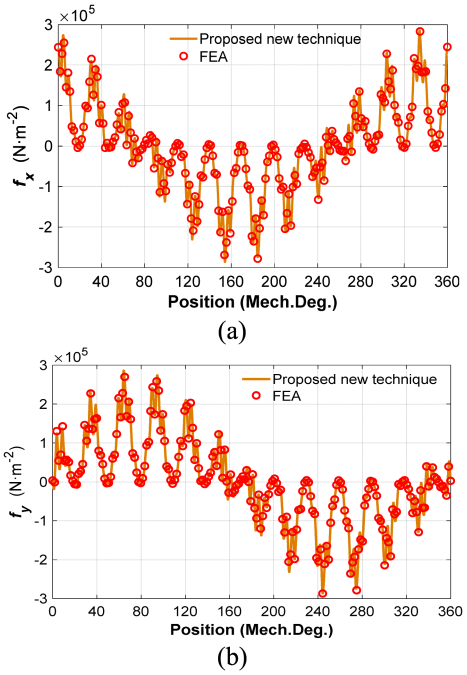


Fig. 14. Loaded magnetic local traction. (a) $x$-direction component. (b) $y$-direction component.

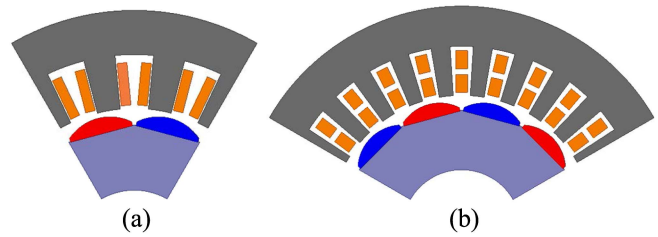


Fig. 16. Fractional slot SPM motor models (one basic section). (a) 18-slot/12-pole SPM motor. (b) 27-slot/12-pole SPM motor.

TABLE IV
SLOT PARAMETERS OF FRACTIONAL SLOT MOTORS

| | 18-slot/12-pole motor | 27-slot/12-pole motor |
|---|---|---|
| Type of slot | radial opening slot | radial opening slot |
| Slot depth | 33 mm | 33 mm |
| Slot width angle | 11° | 7.3° |

fundamental values of the loaded radial flux density between the two methods is 0.57%. It verifies that this new technique also has high accuracy in calculating the loaded air-gap magnetic density, which lays a foundation for accurately solving the electromagnetic torque of the motor. At the same time, the comparison between the results of the FEA and the proposed new technique also indicates that the motor researched in this article has no seriously magnetic saturation, and the assumptions of the subdomain model in Section III have little effect on the accuracy of this new technique.

*5) Magnetic Local Traction and UMF:* According to (33) and (34), the loaded magnetic local traction $f_x$ and $f_y$ can be calculated by the proposed new technique, which are in good agreement with the results of the FEA, as shown in Fig. 14.

Then, Fig. 15 shows the UMFs calculated by the proposed new technique and the FEA when the rotor rotates for one circle. It can be seen that the maximum value of UMF calculated by the proposed new technique is only 0.03 N, and that of FEA is 0.18 N, both of which are very small. Because the motor studied in this article has a symmetrically distributed magnetic field, it has no UMF. The values in Fig. 15 are mainly caused by calculation errors.

*6) Fractional Slot Motors:* The proposed new technique is applicable not only to integral slot motors but also to fractional slot motors because it is applicable to the motor models with arbitrary pole-slot combinations. In Fig. 16, two different types of three-phase fractional slot SPM motors, i.e., 18-slot/12-pole and 27-slot/12-pole SPM motors, are established and analyzed, and the slot parameters of the two motors are shown in Table IV. It is worth noting that the even periodic boundary condition is required for the proposed new technique to calculate the above two fractional slot motors.

Considering the periodical feature of the machine structure (slots and poles), there are 6 basic (unit) sections in the 18-slot/12-pole motor, so the model with a pair of poles is given in Fig. 16(a). In the 27-slot/12-pole motor, there are 3 unit sections so the model with two pairs of poles is given in Fig. 16(b). Then, using the proposed new technique, the model with a pair of poles needs to be calculated for the 18-slot/12-pole motor, and for the 27-slot/12-pole motor, the model with two pairs of poles needs to be calculated. The no-load air-gap flux densities of the above fractional slot SPM motors are calculated by the

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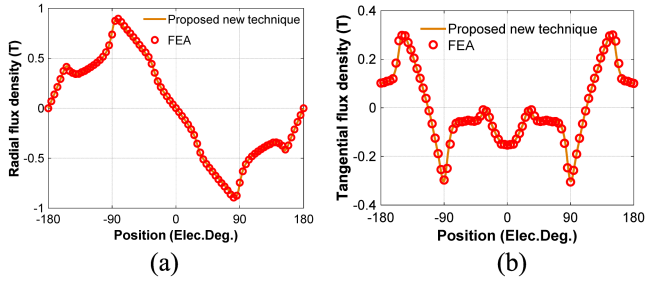


Fig. 17. No-load flux densities of the 18-slot/12-pole SPM motor. (a) Radial component. (b) Tangential component.

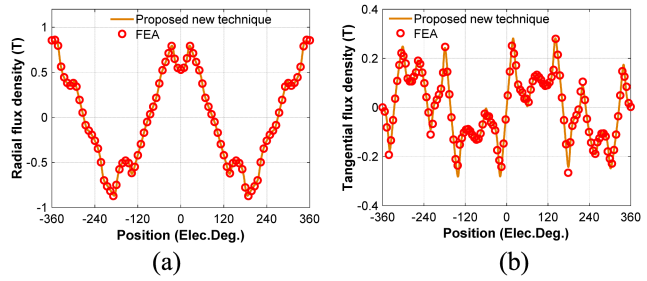


Fig. 18. No-load flux densities of the 27-slot/12-pole SPM motor. (a) Radial component. (b) Tangential component.

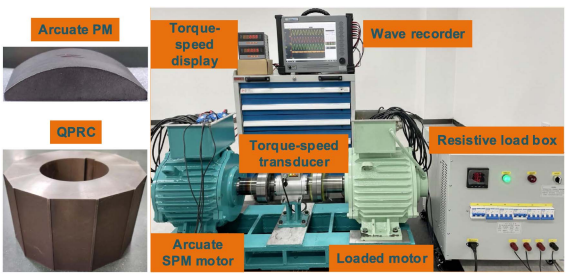


Fig. 19. Prototype structure and experimental platform.

proposed technique and compared with the results of the FEA, as shown in Figs. 17 and 18.

As can be seen from Figs. 17 and 18, the no-load air-gap flux densities of the two fractional slot motors calculated by the proposed new technique are in good agreement with those of the FEA, which shows that the new technique for the subdomain method is also applicable to fractional slot motors.

In addition, the time consumption of the no-load flux density of the 18-slot/12-pole (the unit section is a 3-slot/2-pole model) and the 27-slot/12-pole (the unit section is a 9-slot/4-pole model) motors is 1.14 s and 1.31 s, respectively. It shows that with the increase of the number of slots, the time consumption of the proposed new technique also increases.

## B. Experimental Verification

Using the parameters provided in Table I, a 12-pole/3-phase arcuate SPM prototype is fabricated further to verify the accuracy of the proposed new technique. Fig. 19 shows the arcuate

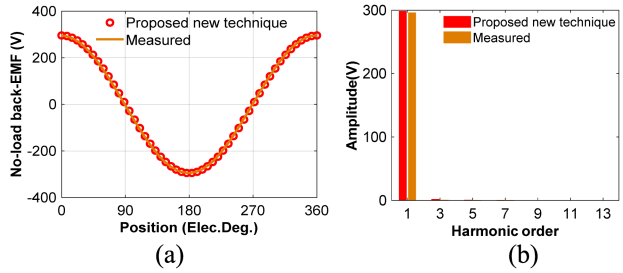


Fig. 20. No-load back-EMF predicted by the proposed new technique together with the measurement of the prototype. (a) Waveforms. (b) Harmonic components.

PM, the QPRC and the experimental platform system of the prototype. In the experimental platform system, a torque-speed transducer connects the arcuate SPM prototype with a loaded motor, and the loaded motor and the resistive load box compose the loaded system.

In the no-load back-EMF test of the prototype, the loaded motor drives the prototype to the rated speed (900 r/min), and the wave recorder can record waveforms of the back-EMF. Fig. 20 shows back-EMF waveforms and their harmonic components obtained by the two methods.

In Fig. 20(a), waveforms of the back-EMF obtained by two methods are in good agreement, and it can be calculated from Fig. 20(b) that the difference of fundamental values between the two methods is only 0.78%, which is mainly due to the machining tolerance of the motor or the measurement error.

In the loaded test of the prototype, the control mode with $I_d = 0$ is used, and the loaded current of the prototype can be adjusted through the resistive load box. The torque-speed display can output the signals of torque. The dynamic torque curve (the blue one) obtained by the experiment under three load conditions (light load, half load and full load) is shown in Fig. 21(a), and the experimental results are in good agreement with those of the proposed new technique. Fig. 21(b) shows curves of average torque with currents $I_q$ obtained by the experiment and the proposed new technique.

As can be seen from Fig. 21(b), under different $I_q$, average torques calculated by the proposed new technique are slightly greater than the measured ones. Because the calculated results are electromagnetic torques, which should be slightly larger than the measured output torques. At the same time, with the increase of $I_q$, the core saturation will also make the experimental results smaller than the analytical ones. In Fig. 21(b), the average error of torques obtained by the two methods is 2.9%, which verifies that the proposed new technique can predict the electromagnetic torque accurately.

## C. Rapidity Analysis of the Proposed New Technique

The rapidity of the proposed new technique with the odd periodic boundary condition is verified by comparing the time consumption with the aperiodic model (the subdomain model without the periodic boundary condition) and the FEA. Using a same computer, Table V shows the time consumption of three

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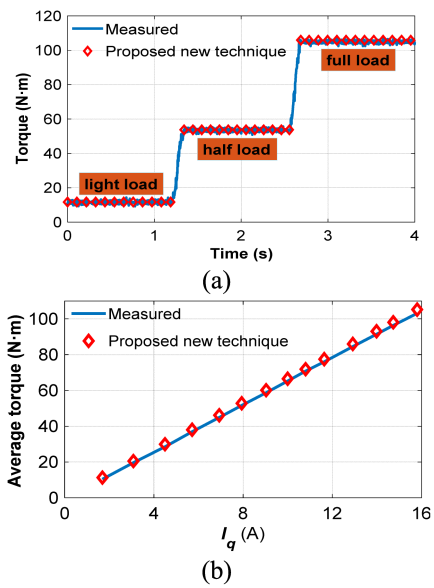


Fig. 21. Comparison of torque. (a) Curves of dynamic torque. (b) Curves of average torque with $I_q$.

TABLE V
COMPARISON OF THE TIME CONSUMPTION

| Performance | Odd periodic | Aperiodic | FEA |
|---|---|---|---|
| No-load flux density | 0.50 s | 3.72 s | 15.7 s |
| Cogging torque | 13.2 s | 51.2 s | 299 s |
| No-load back-EMF | 13.3 s | 51.4 s | 299 s |
| Loaded flux density | 0.51s | 3.74 s | 299 s |
| Winding inductance | 0.06 s | 0.32 s | 299 s |
| Electromagnetic torque | 0.51 s | 3.70 s | 299 s |
| UMF | 13.3 s | 51.5 s | 299 s |

methods when calculating various performances of the integral slot motor. It is worth noting that the no-load flux density is simulated in the static field of the FEA, while others are simulated in the transient field.

As seen from Table V, when the number of segment pairs is $N_z = 16$, the calculation speed of the new technique can also be fast. In most cases, the time consumption of the new technique is two orders of magnitude less than the FEA and nearly three times less than the aperiodic model. In conclusion, the rapidity of the proposed new technique can be verified.

What's more, the time consumption for calculating no-load flux densities of the integral slot motor (a 6-slot/1-pole model is calculated using the odd periodic boundary condition) and the 18-slot/12-pole fractional slot motor (a 3-slot/2-pole model is calculated using the even periodic boundary condition) is 0.50 s and 1.14 s, respectively. It means that the calculation speed of the odd periodic model is faster than that of the even periodic model, although the odd periodic model has more slots, mainly because the dimension of the matrix (as shown in (27)) in the odd periodic model is smaller.

## V. CONCLUSION

In this article, a new technique for the subdomain method is proposed to address the special issues due to the shaped magnet and the QPRC in SPM motors. The proposed new technique uses the magnet symmetrical segmentation method, the idea of variable rotor core radius and the periodic boundary condition to predict various electromagnetic performances of an eccentric-shaping arcuate SPM motor. The accuracy and rapidity of the proposed new technique are verified by the FEA and the experiment. The difference of fundamental values of the loaded radial flux density calculated by the proposed new technique and the FEA is only 0.57%, and the average error of torque obtained by this new technique and the experiment is less than 3%. At the same time, the proposed new technique is applicable not only to integral slot motors, but also to fractional slot motors. Furthermore, the calculation speed of the proposed new technique is two orders of magnitude faster than that of the FEA. Thus, this new technique can play an essential role in the initial design and optimization of arcuate SPM motors with shaped magnets and a QPRC. In the future, the actual permeability of the stator and rotor cores should be considered to improve the accuracy of the calculation further.

## APPENDIX

There are nine equations about the ten unknown coefficients.

1) *The equations of $A_{m1}$ and $C_{m1}$.*

$$A_{m1} = A_{m2}\frac{R_{u(j)}}{mc}\frac{P_{mc}\left(R_{u(j)}, R_s\right)}{E_{mc}\left(R_{u(j)}, R_s\right)} + B_{m2}\frac{R_s}{mc}\frac{2}{E_{mc}\left(R_s, R_{u(j)}\right)}$$

$$C_{m1} = C_{m2}\frac{R_{u(j)}}{mc}\frac{P_{mc}\left(R_{u(j)}, R_s\right)}{E_{mc}\left(R_{u(j)}, R_s\right)} + D_{m2}\frac{R_s}{mc}\frac{2}{E_{mc}\left(R_s, R_{u(j)}\right)}$$

2) *The equations of $A_{m2}$ and $C_{m2}$.*

$$A_{m2} = A_{m1}\frac{mc}{\mu_r R_{u(j)}}\frac{E_{mc}\left(R_{u(j)}, R_{l(j)}\right)}{P_{mc}\left(R_{u(j)}, R_{l(j)}\right)} - \frac{1}{\mu_r}X'_{mj}\left(R_{u(j)}\right)\sin\left(mc\Delta\right)$$

$$C_{m2} = C_{m1}\frac{mc}{\mu_r R_{u(j)}}\frac{E_{mc}\left(R_{u(j)}, R_{l(j)}\right)}{P_{mc}\left(R_{u(j)}, R_{l(j)}\right)} + \frac{1}{\mu_r}X'_{mj}\left(R_{u(j)}\right)\cos\left(mc\Delta\right)$$

where $X'_{mj}(r)$ is the derivative of $X_{mj}(r)$ with respect to $r$.

3) *The equations of $B_{m2}$ and $D_{m2}$.*

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If the odd periodic boundary condition is used in the proposed new technique, $B_{m2}$ and $D_{m2}$ can be given by

$$B_{m2} = \sum_{i=1}^{Z/2c} 2cK_c f(m,i) + \sum_{i=1}^{Z/2c}\sum_{n=1}^{\infty} A_{n3i}\frac{G_n}{R_s}\frac{E_{G_n}(R_s,R_{sa})}{P_{G_n}(R_s,R_{sa})}\frac{2cf(n,m,i)}{\pi}$$

$$D_{m2} = \sum_{i=1}^{Z/2c} 2cK_c g(m,i) + \sum_{i=1}^{Z/2c}\sum_{n=1}^{\infty} A_{n3i}\frac{G_n}{R_s}\frac{E_{G_n}(R_s,R_{sa})}{P_{G_n}(R_s,R_{sa})}\frac{2cg(n,m,i)}{\pi}$$

If the even periodic boundary condition is used in the proposed new technique, $B_{m2}$ and $D_{m2}$ are rewritten as

$$B_{m2} = \sum_{i=1}^{Z/c} cK_c f(m,i) + \sum_{i=1}^{Z/c}\sum_{n=1}^{\infty} A_{n3i}\frac{G_n}{R_s}\frac{E_{G_n}(R_s,R_{sa})}{P_{G_n}(R_s,R_{sa})}\frac{cf(n,m,i)}{\pi}$$

$$D_{m2} = \sum_{i=1}^{Z/c} cK_c g(m,i) + \sum_{i=1}^{Z/c}\sum_{n=1}^{\infty} A_{n3i}\frac{G_n}{R_s}\frac{E_{G_n}(R_s,R_{sa})}{P_{G_n}(R_s,R_{sa})}\frac{cg(n,m,i)}{\pi}$$

Then,

$$K_c = -\frac{\mu_0 J_{ui}}{2\pi}R_s + \left[\frac{\mu_0 R_{sc}^2}{2\pi}(J_{ui}-J_{di}) + \frac{\mu_0 R_{sa}^2}{2\pi}J_{di}\right]\frac{1}{R_s}$$

$$f(m,i) = \frac{1}{mc}\left\{\sin\left[mc\left(\theta_i+\frac{b_s}{2}\right)\right] - \sin\left[mc\left(\theta_i-\frac{b_s}{2}\right)\right]\right\}$$

$$g(m,i) = -\frac{1}{mc}\left\{\cos\left[mc\left(\theta_i+\frac{b_s}{2}\right)\right] - \cos\left[mc\left(\theta_i-\frac{b_s}{2}\right)\right]\right\}$$

When $mc \neq G_n$

$$f(n,m,i) = \frac{mc\{\sin[mc(\theta_i - \frac{b_s}{2})] - (-1)^n \sin[mc(\theta_i + \frac{b_s}{2})]\}}{G_n^2 - (mc)^2}$$

$$g(n,m,i) = \frac{mc\{(-1)^n \cos[mc(\theta_i + \frac{b_s}{2})] - \cos\left[mc\left(\theta_i - \frac{b_s}{2}\right)\right]\}}{G_n^2 - (mc)^2}$$

When $mc = G_n$

$$f(n,m,i) = \frac{b_s}{2}\cos\left[mc\left(\theta_i - \frac{b_s}{2}\right)\right]$$

$$g(n,m,i) = \frac{b_s}{2}\sin\left[mc\left(\theta_i - \frac{b_s}{2}\right)\right]$$

4) *The equations of $A_{n3i}$ and $A_{n4i}$.*

$$A_{n3i} = A_{n4i} = \sum_{m=1}^{\infty}\left[A_{m2}\frac{R_{u(j)}}{mc}\frac{2}{E_{mc}(R_{u(j)},R_s)} + B_{m2}\frac{R_s}{mc}\frac{P_{mc}(R_s,R_{u(j)})}{E_{mc}(R_s,R_{u(j)})}\right]\frac{2f(n,m,i)}{b_s} + \sum_{m=1}^{\infty}\left[C_{m2}\frac{R_{u(j)}}{mc}\frac{2}{E_{mc}(R_{u(j)},R_s)} + D_{m2}\frac{R_s}{mc}\frac{P_{mc}(R_s,R_{u(j)})}{E_{mc}(R_s,R_{u(j)})}\right]\frac{2g(n,m,i)}{b_s}$$

5) *The equations of $B_{03i}$ and $B_{04i}$.*

$$B_{03i} = \sum_{m=1}^{\infty}\left[A_{m2}\frac{R_{u(j)}}{mc}\frac{2}{E_{mc}(R_{u(j)},R_s)} + B_{m2}\frac{R_s}{mc}\frac{P_{mc}(R_s,R_{u(j)})}{E_{mc}(R_s,R_{u(j)})}\right]\frac{f(m,i)}{b_s} + \sum_{m=1}^{\infty}\left[C_{m2}\frac{R_{u(j)}}{mc}\frac{2}{E_{mc}(R_{u(j)},R_s)} + D_{m2}\frac{R_s}{mc}\frac{P_{mc}(R_s,R_{u(j)})}{E_{mc}(R_s,R_{u(j)})}\right]\frac{g(m,i)}{b_s}$$

$$B_{04i} = B_{03i} + \frac{\mu_0 J_{ui}}{4}\left(R_s^2 - R_{sc}^2\right) + \left[\frac{\mu_0 R_{sc}^2}{2}(J_{ui}-J_{di}) + \frac{\mu_0 R_{sa}^2}{2}J_{di}\right]\ln\frac{R_{sc}}{R_s}$$

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